Powering SIESTA's Electronic Structure Calculations with the SIESTA ToolBox

Part 2 — Advanced STB Functions and Workflows

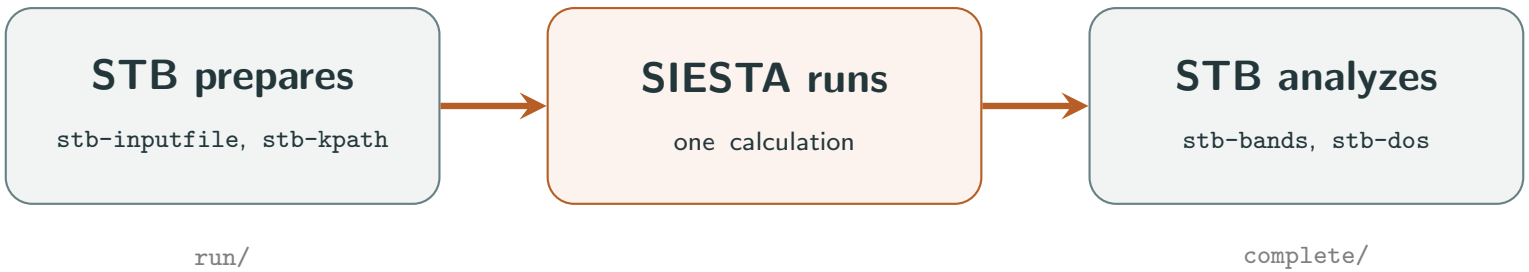
Dr. Carlos Maciel de Oliveira Bastos

International Center of Physics
University of Brasilia

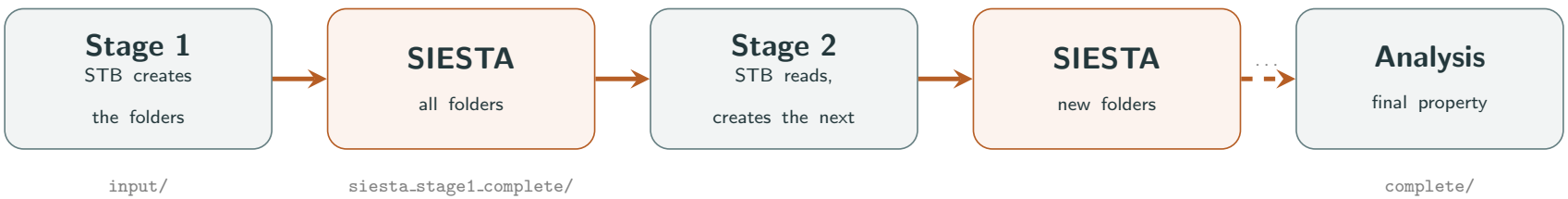


Where We Stopped

Single tool



Workflow



Bands and Density of States

Bloch Electrons and the Brillouin Zone

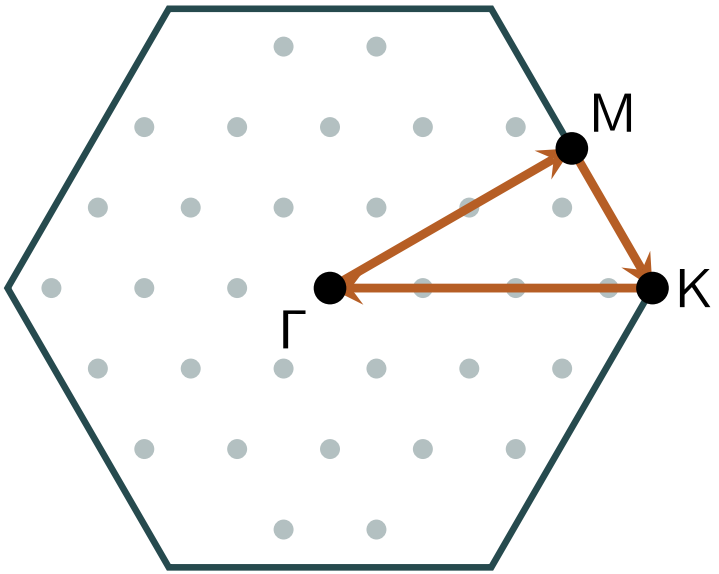
Periodicity labels every state by k

$$\psi_{n\mathbf{k}}(\mathbf{r}) = e^{i\mathbf{k}\cdot\mathbf{r}}\, u_{n\mathbf{k}}(\mathbf{r})$$
$$u_{n\mathbf{k}}(\mathbf{r}+\mathbf{R}) = u_{n\mathbf{k}}(\mathbf{r})$$

- **k** lives in the first Brillouin zone (BZ)
- For each **k**, Kohn–Sham gives a ladder of levels $\varepsilon_n(\mathbf{k})$
- A **band** is $\varepsilon_n(\mathbf{k})$ followed continuously in **k**

Two different uses of k

SCF mesh: a uniform grid that integrates the density.
Band path: lines through high-symmetry points, for plotting.

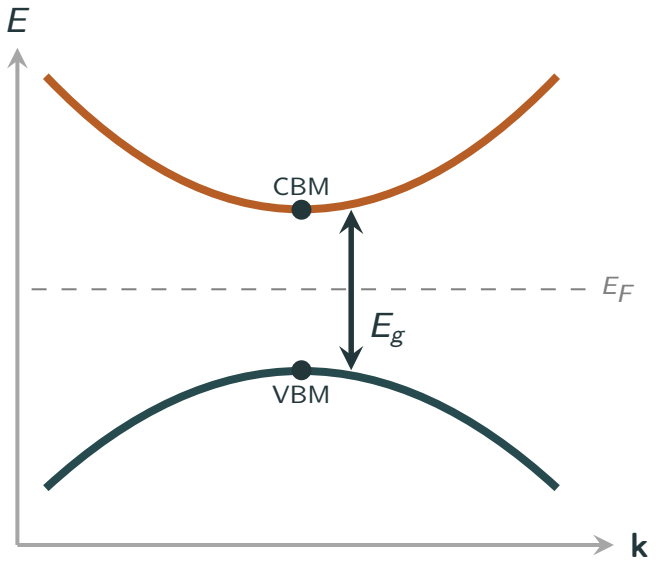


2D hexagonal BZ (h-BN)

- SCF mesh (15×15×1, schematic) — band path Γ–M–K–Γ

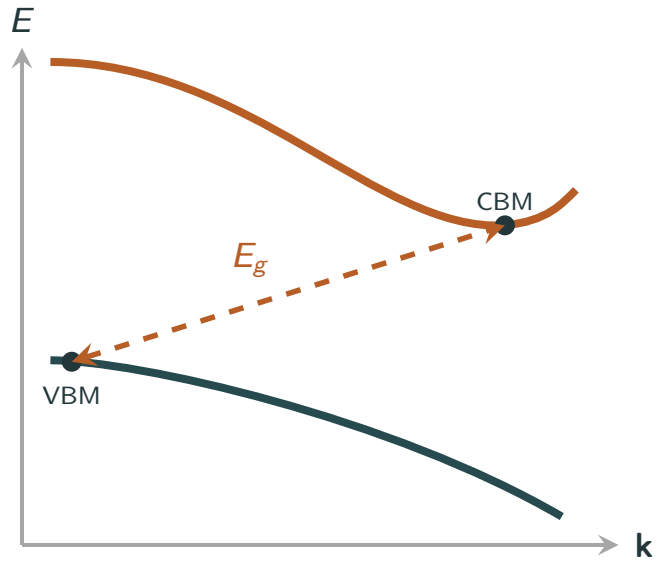
In 2D: one k point along z, a path in the plane.

Reading a Band Structure



Direct gap

VBM and CBM at the same **k**



Indirect gap

different **k**: absorption needs a phonon

What the diagram tells you

- Metal or insulator; size and type of E_g
- Band width: localized vs delocalized states
- Curvature: $1/m^* = \hbar^{-2}\,\partial^2 E/\partial k^2$

Kohn–Sham gap $\neq$ optical gap

- PBE underestimates E_g
- GaAs: 0.36 eV (PBE) vs 1.42 eV (exp.)
- Trust trends and band character more than the number

Density of States and Orbital Projections

Count states per energy

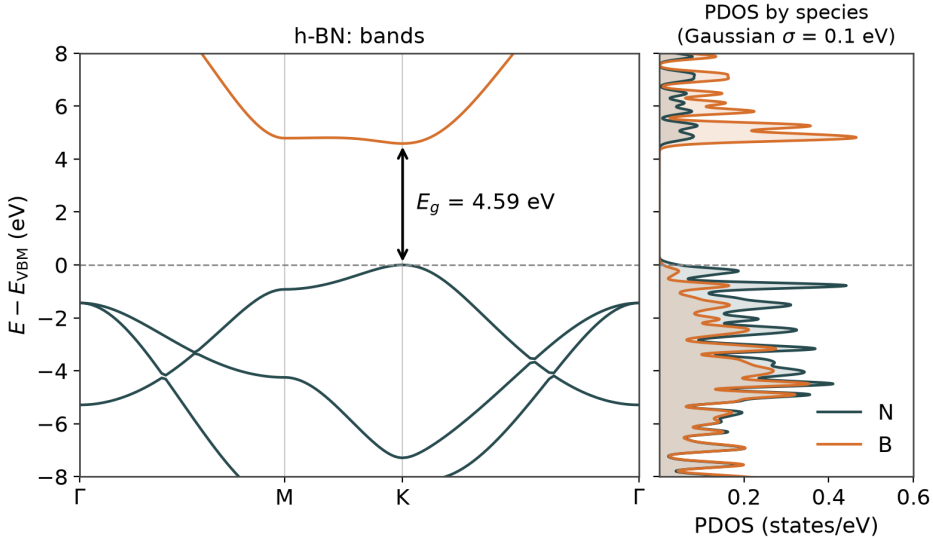
$$g(E) = \sum_n \sum_{\mathbf{k}} w_{\mathbf{k}}\, \delta(E - \varepsilon_{n\mathbf{k}})$$

The δ is replaced by a peak of finite width.

Project onto atomic orbitals μ

$$g_\mu(E) = \sum_{n\mathbf{k}} w_{\mathbf{k}}\, \mathrm{Re}\left[c^*_{\mu,n\mathbf{k}}\, (\mathbf{S}\, \mathbf{c}_{n\mathbf{k}})_\mu\right] \times \delta(E - \varepsilon_{n\mathbf{k}})$$

Natural in LCAO: SIESTA already works in the orbital basis. $\sum_\mu g_\mu = g$.



h-BN (PBE/DZP): valence band mostly N, conduction band mostly B.

DOS needs the whole BZ, not the path

- Integrated on its own uniform grid, `PDOS.kgrid_Monkhorst_Pack`, usually denser than the SCF mesh
- DOS peaks come from flat regions of $\varepsilon_n(\mathbf{k})$

Bands and DOS with STB



What `-t bands` adds to `calc.fdf`

```
%include kpath_bs.fdf
%block ProjectedDensityOfStates
  -20.0 20.0 0.01 8000 eV
%endblock ProjectedDensityOfStates
%PDOS.kgrid_Monkhorst_Pack ...
```

One run, two products

- Path eigenvalues: `siesta.bands`
- Projected DOS: `siesta.PDOS.xml`
- Both shifted to the same VBM

stb-kpath: The High-Symmetry Path

What it does

- Detects 2D/3D: an empty span > 10 Å is a vacuum axis
- Finds the Bravais lattice (ASE)
- Writes the Setyawan–Curtarolo path as a `BandLines` block

Command and h-BN report

```
stb-kpath -f structure.fdf --save-report
# -> kpath_bs.fdf (BandLines)

Dimensionality  : 2D
Bravais lattice : HEX2D
Path            : Γ-M-K-Γ
```

Pitfall: noise breaks symmetry

Relaxed GaAs is cubic only to $\sim 10^{-3}$ Å.

Default `-p 0.0002` $\rightarrow$ **rhombohedral**
`Γ-L-B1-B-Z-Γ-X-Q-F-P1-Z-L-P`

Our example `-p 0.01` $\rightarrow$ **FCC**
`Γ-X-W-K-Γ-L-U-W-L-K | U-X`

Before running SIESTA

Check the reported lattice and space group (GaAs: F$\bar{4}$3m, No. 216).

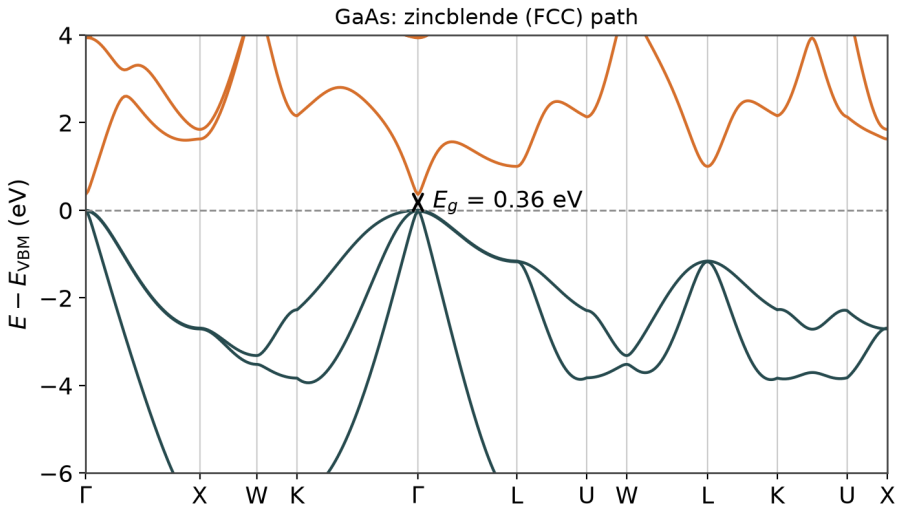
`stb-bands`: Gap, Character, and Plot

Command

```
stb-bands --label siesta \
  --shift vbm --save-report
```

What it reports

- VBM, CBM, direct and indirect gap
- Metal if gap < 0.01 eV (`--gap-tol`)
- **Path vs mesh**: did the path miss the band edges?
- `bands_gnuplot.dat` + `.gplot`



	GaAs	h-BN
Gap, path / mesh (eV)	0.361 / 0.361	4.589 / 4.589
Character	direct (Γ)	direct (K)

Choosing the zero: `--shift`

`vbm`: semiconductors `fermi`: metals

`manual`: common reference across systems

stb-dos: Total, Atom, and Species PDOS

Command

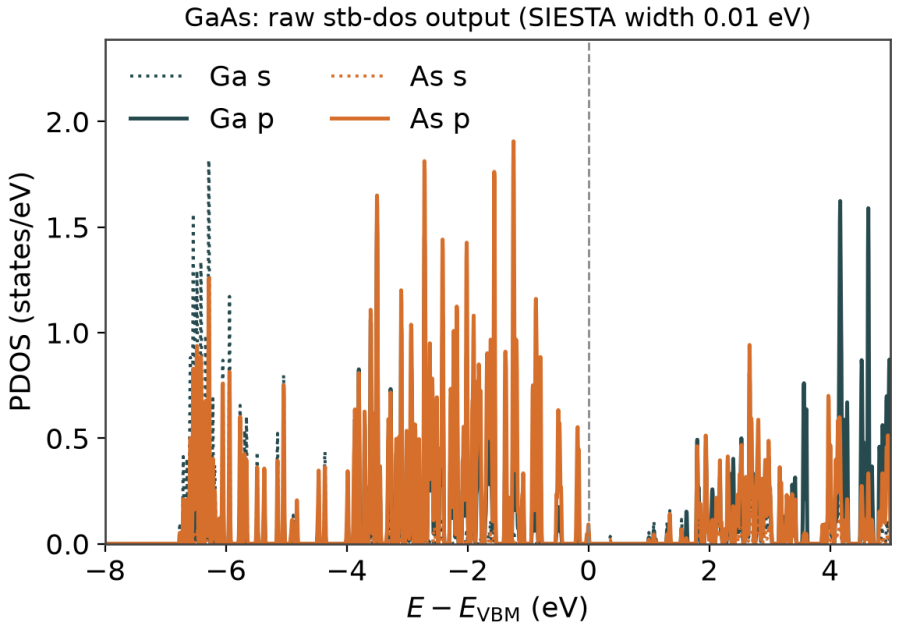
```
stb-dos --label siesta \
  --shift vbm --save-report \
  --save-gnuplot
```

Options

- `--type`: `total`, `atom`, `species`
- `--projection`: `l` (s, p, d), `ml` (p_x, d_{xy}, ...)
- `--shift vbm`: same zero as `stb-bands`

Outputs (each with a `.gplot`)

`dos_total.dat`, `dos_per_atom/`, `dos_per_species/`



Spiky is not wrong

- 0.01 eV peak width: each k point shows
- Smooth it with `stb-convdos` (2.3)

Hands-on: Bands and DOS of GaAs and h-BN

Critical Reading of Electronic Results

Broadening: Resolution, Not Accuracy

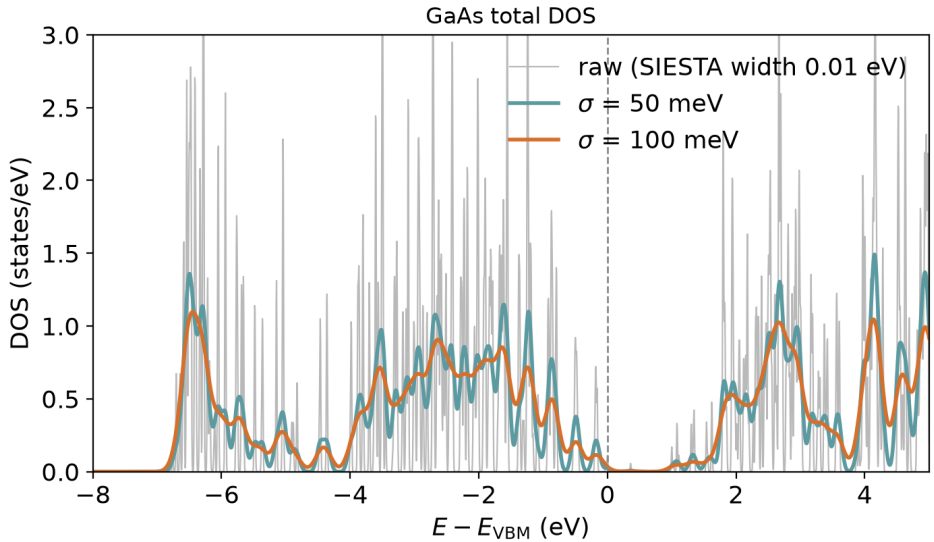
Gaussian convolution

$$\tilde{g}(E) = \int g(E') \, \frac{e^{-(E-E')^2/2\sigma^2}}{\sigma\sqrt{2\pi}} \, dE'$$

$\mathrm{FWHM} = 2.3548\,\sigma$. The kernel has unit area, so the number of states is conserved.

Why the raw DOS is spiky

A finite k mesh gives a finite set of $\varepsilon_{n\mathbf{k}}$. A narrow peak shows each one; a wider one interpolates between them.



Same SIESTA run, three widths.

A smoother curve is not a better calculation

- σ is a display choice: state it in every figure
- Too wide merges peaks and smears the band edges into the gap
- Only a denser k mesh adds information

Work Function and the Vacuum Level

Definition

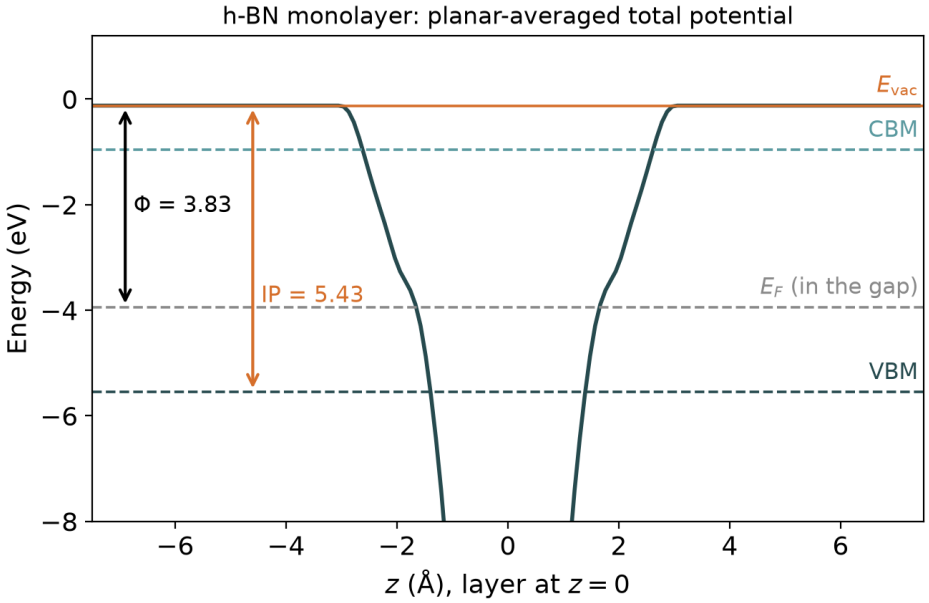
$$\Phi = E_{\text{vac}} - E_F$$

$$\bar{V}(z) = \frac{1}{A}\iint_A V(x, y, z)\, dx\, dy$$

E_{vac} is the flat plateau of the planar-averaged total potential $\bar{V}(z)$ far from the layer.

Requirements

- A vacuum region wide enough to flatten $\bar{V}$
- Dipole correction for asymmetric slabs
- **Not defined** for a periodic 3D crystal (GaAs)



Insulators: E_F floats in the gap

- In h-BN, E_F sits mid-gap and moves with smearing
- Report band-edge quantities: IP $= E_{\text{vac}} - E_{\text{VBM}} = 5.43$ eV, EA $= E_{\text{vac}} - E_{\text{CBM}} = 0.84$ eV

Orbital Character Along the Bands: Fatbands

Weight of orbital μ in state $n\mathbf{k}$

$$w_{\mu,n\mathbf{k}} = \mathrm{Re}\left[c^{*}_{\mu,n\mathbf{k}}\,(\mathbf{S}\,\mathbf{c}_{n\mathbf{k}})_{\mu}\right]$$

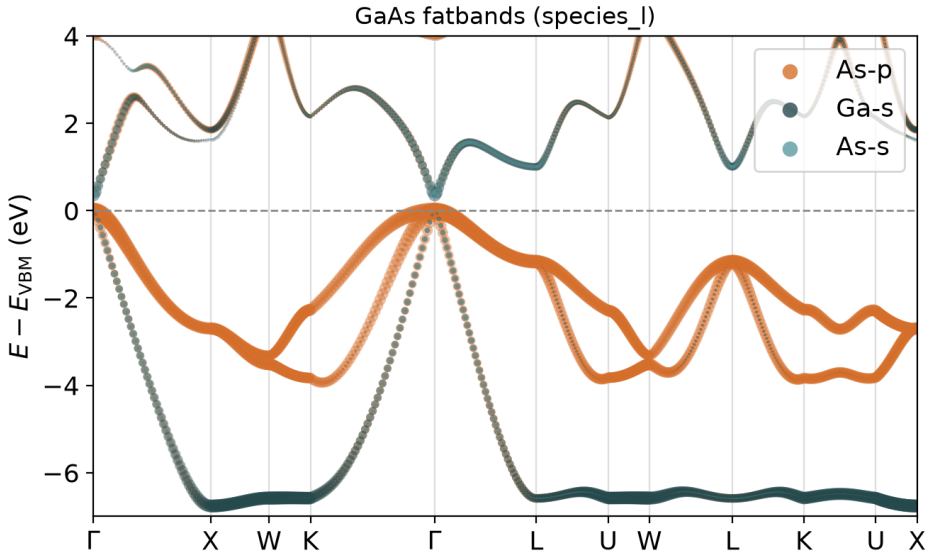
$$\sum_{\mu} w_{\mu,n\mathbf{k}} = 1$$

Marker size $\propto w$. Summing $w\,\delta(E-\varepsilon_{n\mathbf{k}})$ over the whole BZ gives the PDOS.

PDOS vs fatbands

PDOS: *which energies*.

Fatbands: *which energies and which* **k**.

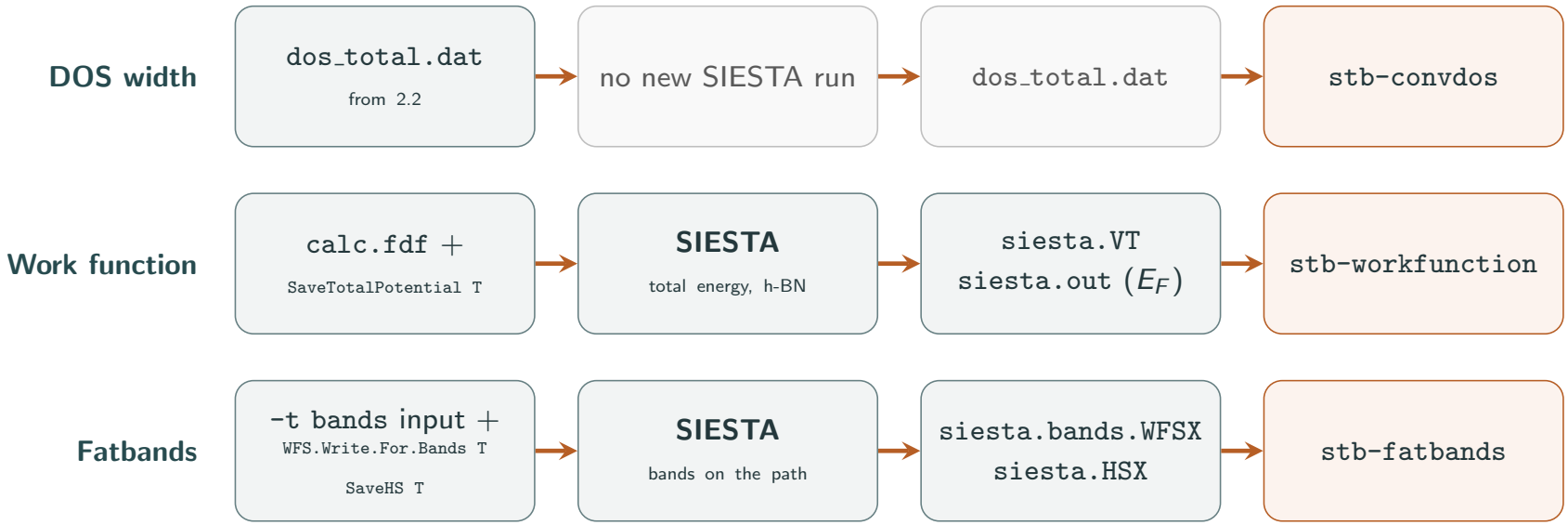


GaAs: VBM is As-p; CBM at Γ is Ga-s/As-s.

A projection, not a measurement

- Mulliken-type weights depend on the basis
- Non-orthogonal basis: a weight can be slightly < 0 or > 1
- Compare trends, not decimals

Three Readings, Three Requirements



SIESTA only writes what you ask for

- `.VT`, `.WFSX`, `.HSX` are off by default: set the flag *before* the run
- A small wrapper `.fdf` that `%includes calc.fdf` keeps the template untouched

stb-convdos: Gaussian Broadening

```
stb-convdos --file dos_total.dat --sigma 50 \
  --out dos_total_50meV.dat --no-plot --save-report
```

Options

- `--sigma` or `--fwhm`, in meV
- `--dir`: broaden a whole `stb-dos` output tree
- `--size`: kernel width in samples (default $\pm 3\sigma$)
- Every DOS column is broadened; the energy grid is kept

GaAs report, $\sigma = 50$ meV

```
Grid spacing :  0.005 eV
Sigma        :  50 meV (10 samples)
Kernel size  :  61 samples
```

Conservation check

Column	Before	After
s	1.8251	1.8250
p	7.5399	7.5311
d	10.8686	10.8667

Read the conservation check

- Drift above a few tenths of a percent: weight leaked off the energy window
- Fix: widen the PDOS window in SIESTA, or reduce σ

stb-workfunction: Vacuum Level from .VT

```
stb-workfunction --label siesta \
  --save-report --save-gnuplot
```

What it does

- Detects the vacuum axis from the structure
- Averages .VT over the plane; finds flat plateaus
- Asymmetric slab: reports **two** work functions
- Warns if the plateau has a slope (missing dipole correction)

h-BN report

```
E_F          : -3.9494 eV (siesta.out)
Axis         :  2 (z), auto-detected
Plateau std :  0.000014 eV (flat)

Vacuum level:  -0.1240 eV
Work function:  3.8254 eV
```

Check before quoting

Flat plateau? Symmetric slab? Metal or insulator? For h-BN, quote IP = 5.43 eV next to Φ.

Do not run it on GaAs: a fully periodic 3D cell has no vacuum, so there is no E_{vac}.

stb-fatbands: Orbital-Projected Bands

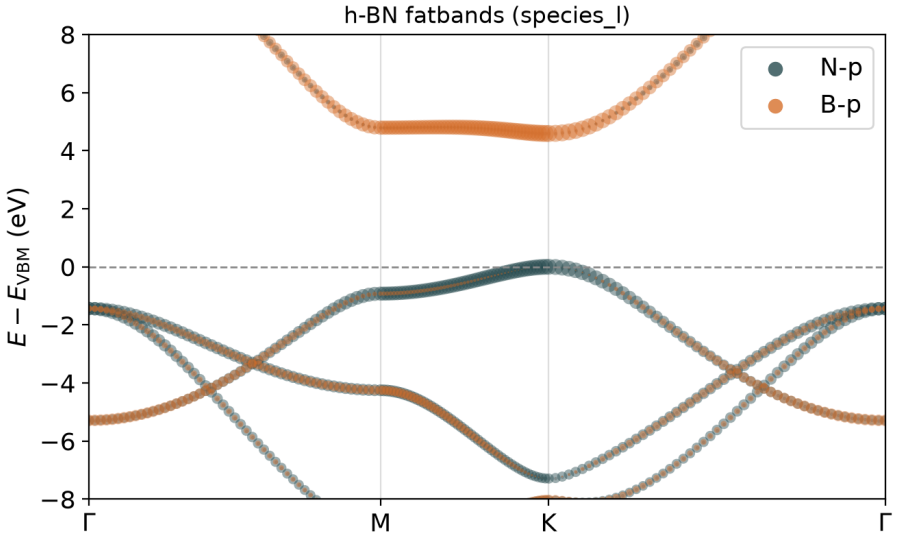
```
stb-fatbands --label siesta --shift vbm \
  --projection species_l \
  --save-report --save-gnuplot
```

Projections

l, ml, atom, species, species_l (e.g. N-p, B-s); --category keeps only some

Built-in checks

- .bands and .WFSX cross-checked at sampled k points
- With .HSX: overlap-weighted (accurate) weights
- Without it: falls back to the .fdf, with a warning



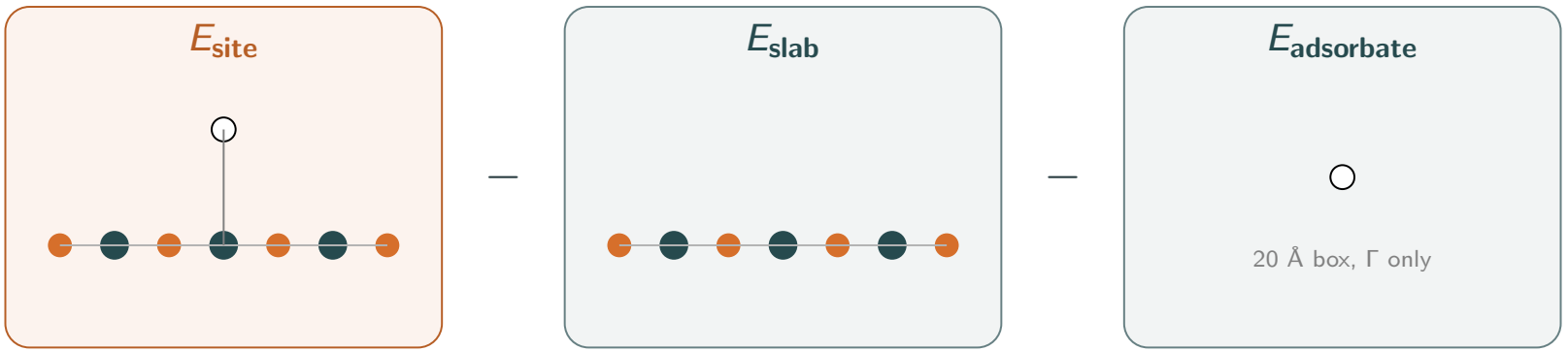
h-BN

- At K: valence N-*p*, conduction B-*p*
- Same picture as the PDOS, now resolved in **k**

Hands-on: Broadening, Work Function, and Fatbands

Adsorption on h-BN

Adsorption Energy: Three Calculations, One Number



$$E_{\text{ads}} = E_{\text{site}} - E_{\text{slab}} - E_{\text{adsorbate}} \qquad (E_{\text{ads}} < 0 \Rightarrow \text{favorable})$$

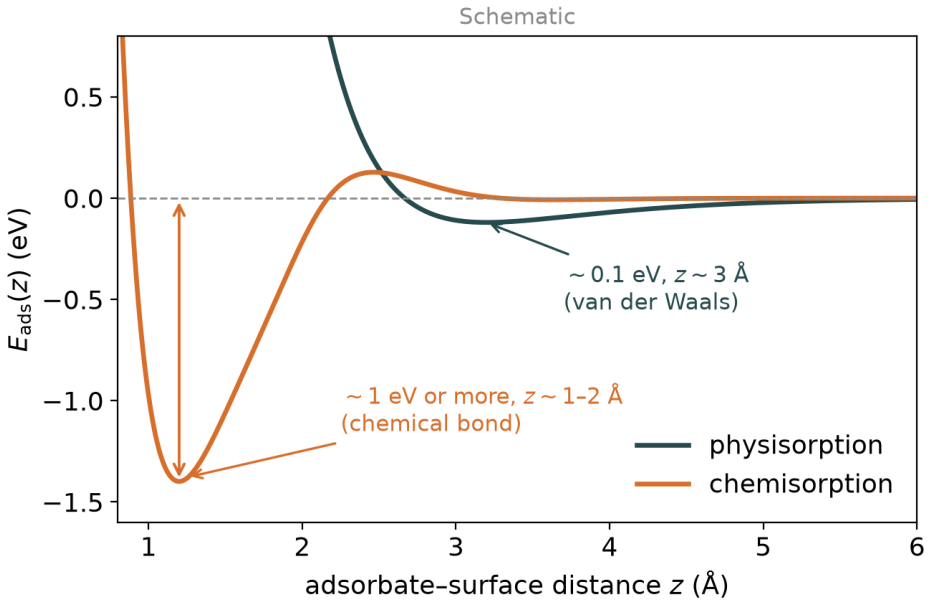
Same settings everywhere

Basis, mesh, functional, D3, and pseudopotentials must match; otherwise the subtraction does not cancel numerical errors.

Which reference for H?

Atomic H measures the bond to the surface. $\frac{1}{2}E_{H_2}$ asks whether H prefers the surface or H_2 (H_2 binding: 4.5 eV, about 2.2 eV per H).

Physisorption or Chemisorption?



Physisorption

Dispersion only; electronic structure barely changes. Weak, long distance: needs D3.

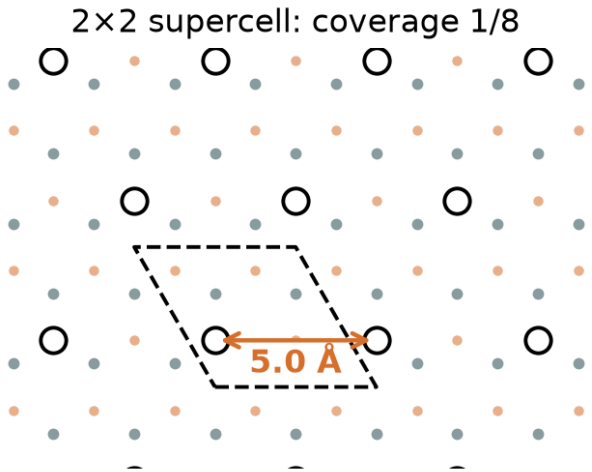
Chemisorption

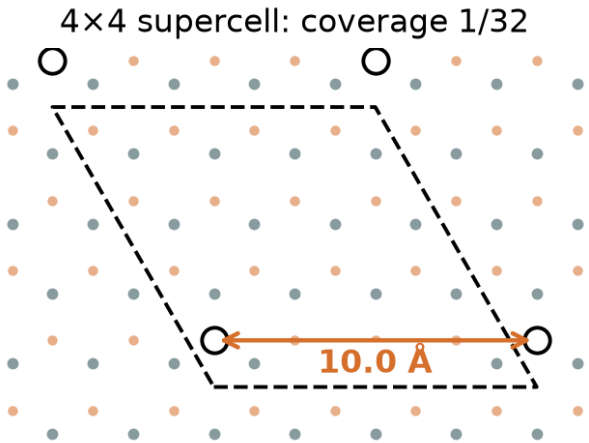
A bond forms: charge transfer, rehybridization, possible barrier. On h-BN, H on top of B or N pulls the atom out of plane ($sp^2 \rightarrow sp^3$-like).

The start height decides which well you find

- A relaxation only goes downhill: 1.5 Å targets the chemical well, a high start can end in the shallow one
- Check the final z and the bonds

Supercell, Coverage, and Vacuum





In-plane

Periodic images interact. Bigger cell → lower coverage, closer to an isolated adsorbate.

Out-of-plane

15 Å cell: enough vacuum for the potential to flatten. Dipole correction removes the artificial field.

k points

A 4×4 cell needs 4× fewer k per direction: 3×3×1 here ≈ 12×12 in the primitive cell.

Basis-Set Superposition Error in Adsorption

Why LCAO over-binds

In the site calculation, H can borrow the slab's orbitals (and vice versa). Each fragment is described better *together* than *alone*, so E_{ads} comes out too negative.

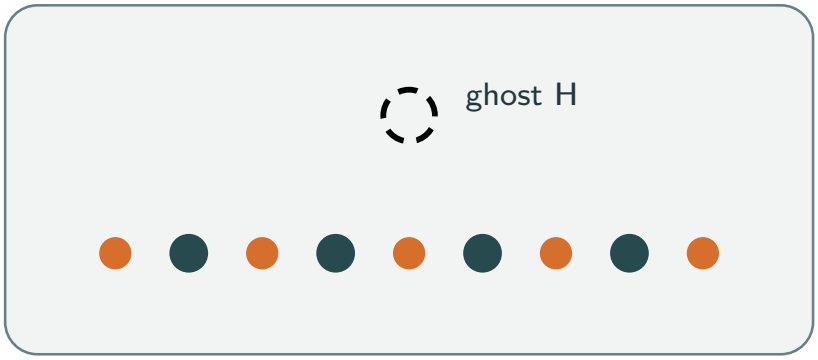
Counterpoise (Boys–Bernardi)

$$E_{\text{ads}}^{\text{CP}} = E_{\text{ads}} + \delta_{\text{slab}} + \delta_{\text{H}}$$

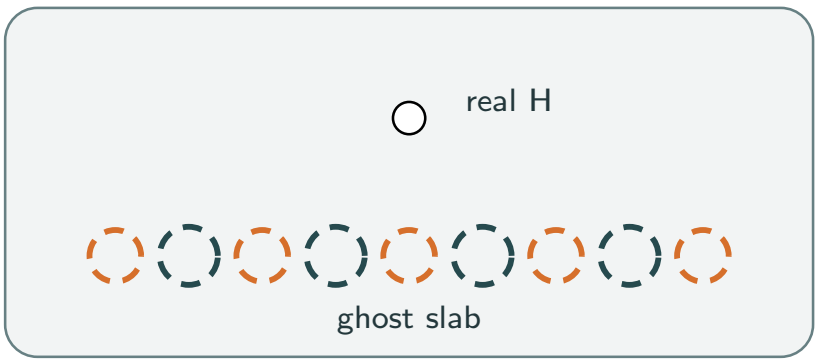
$$\delta_X = E_X - E_X^{[\text{ghost}]} > 0$$

Ghost = orbitals without nucleus or electrons. Both terms of δ_X use fragment X at the **relaxed site geometry**.

`bsse_slab`



`bsse_adsorbate`



Same correction you saw for cohesive energy

- Part 1: BSSE moved E_{coh} of GaAs by $\sim$1 eV/atom
- Adsorption: it can change the ranking of close sites

Getting the Physics Right Before the Numbers

SPIN

Atomic H is a doublet ($1\,\mu_B$)

Isolated H and every site run spin-polarized

DISPERSION

Grimme DFT-D3

PBE alone misses van der Waals attraction

DIPOLE

`Slab.DipoleCorrection`

One-sided adsorption creates a dipole across the vacuum

CELL

`MD.VariableCell false`

Relax atoms only; keep the vacuum

Sites are starting points

ontop, *bridge*, and *hollow* label where the adsorbate *starts*. The relaxation may move it elsewhere: always inspect the final geometry.

First occurrence wins

- SIESTA reads the *first* value of each label
- STB puts these settings in `config_extra.fdf`, included at the **top** of `calc.fdf`

Beyond E_{ads}: Gibbs Free Energy of Adsorption

Add vibrations and entropy

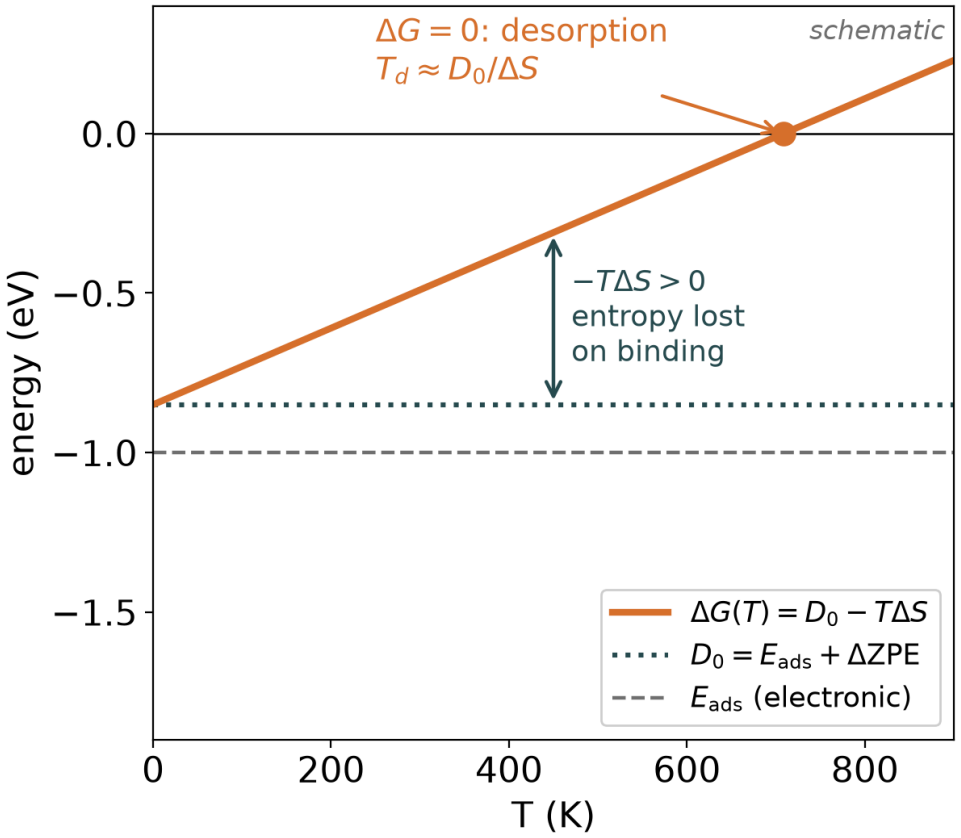
$$\Delta G(T) = \underbrace{E_{ads} + \Delta \mathrm{ZPE}}_{D_0} - T\,\Delta S$$

$\mathrm{ZPE} = \sum_i \frac{1}{2}\hbar\omega_i$, ω_i from a finite-difference Hessian (harmonic approximation).

$\Delta X = X_{site} - X_{reference}$.

H on h-BN

- **Site**: 3 H modes (1 stretch, 2 bends), substrate frozen
- **Reference**: free H, no vibrations, but translational entropy (ideal gas, 1 bar)



Binding costs entropy

- The adsorbate loses freedom: $\Delta S < 0$, so ΔG rises with T
- T_d is a thermodynamic estimate, not a kinetic prediction

How ΔS Is Computed

$$\Delta S(T) = S_{\text{site}}(T) - S_{\text{ref}}(T)$$

Adsorbed H: harmonic oscillators

$$S_{\text{site}} = k_B \sum_{i=1}^{3} \left[\frac{x_i}{e^{x_i} - 1} - \ln\left(1 - e^{-x_i}\right) \right]$$

$$x_i = \frac{\hbar \omega_i}{k_B T}$$

ω_i from the local Hessian. Stiff modes ($\hbar\omega \gg k_B T$) contribute almost nothing.

$\tilde{\nu}$ (cm^{-1})	500	1000	2500
S at 298 K (J/(mol K))	2.8	0.4	0.0

Free H atom: ideal gas at 1 bar

$$S_{\text{ref}} = S_{\text{trans}} + S_{\text{rot}} + S_{\text{vib}} + S_{\text{elec}}$$

$$S_{\text{trans}} = k_B \left[\ln\left(\left(\frac{2\pi m k_B T}{h^2} \right)^{3/2} \frac{k_B T}{p^\circ} \right) + \frac{5}{2} \right]$$

Atom: $S_{\text{rot}} = S_{\text{vib}} = 0$. $S_{\text{elec}} = k_B \ln(2s + 1)$, s from the SIESTA spin moment.

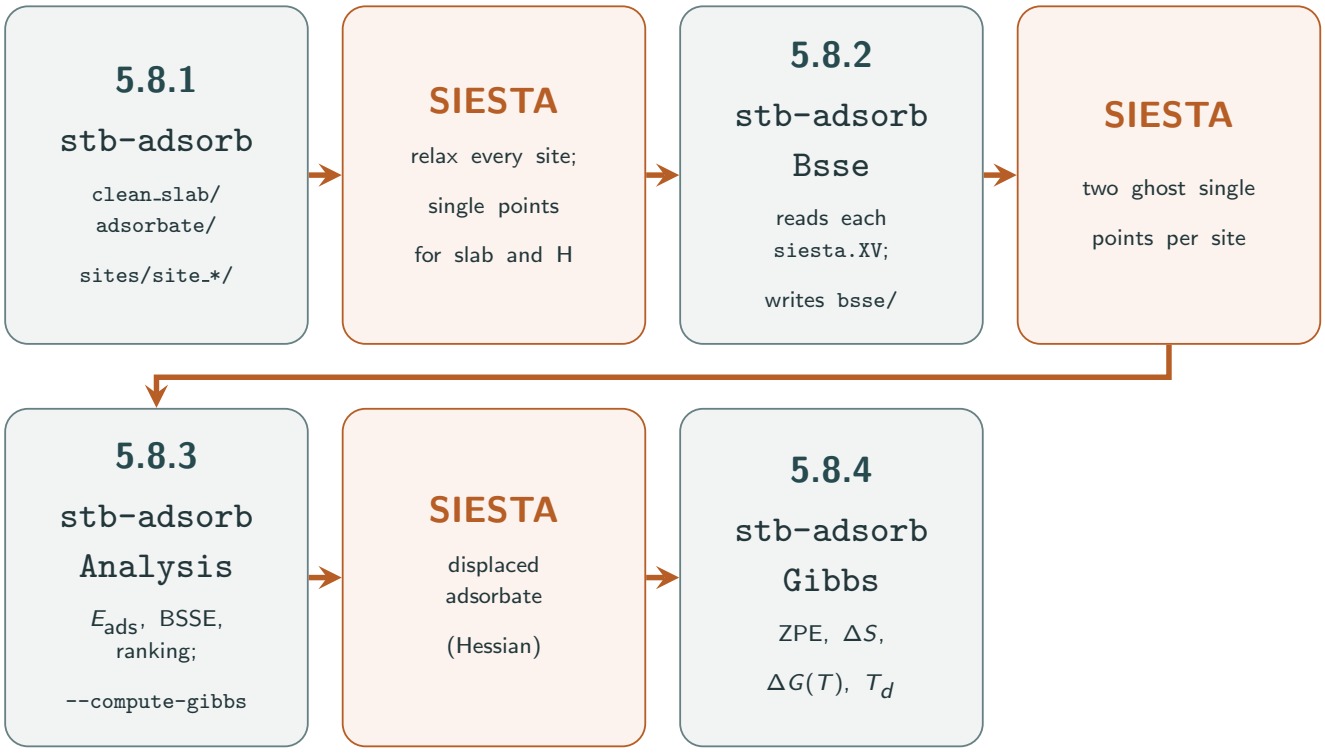
H at 298 K: $108.95 + 5.76 = \mathbf{114.72}$ J/(mol K)

(NIST-JANAF: 114.72)

Translation dominates

- $\Delta S \approx -110$ J/(mol K): $-T\Delta S \approx +0.34$ eV at 300 K
- Almost all of it is the lost translation of the gas-phase atom (ASE `IdealGasThermo`)

Workflow 5.8: Adsorption with STB



Starting point

```
stb-supercell -f structure.fdf -d 4 4 1 \
  -o structure.fdf
stb-inputfile structure.fdf -t relax -p dojo
```

then set `kgrid.MonkhorstPack [3 3 1]`

Why BSSE is a separate stage

- Ghost runs need the *relaxed* site geometry
- It exists only after the first SIESTA round

stb-adsorb: Sites and Reference Folders

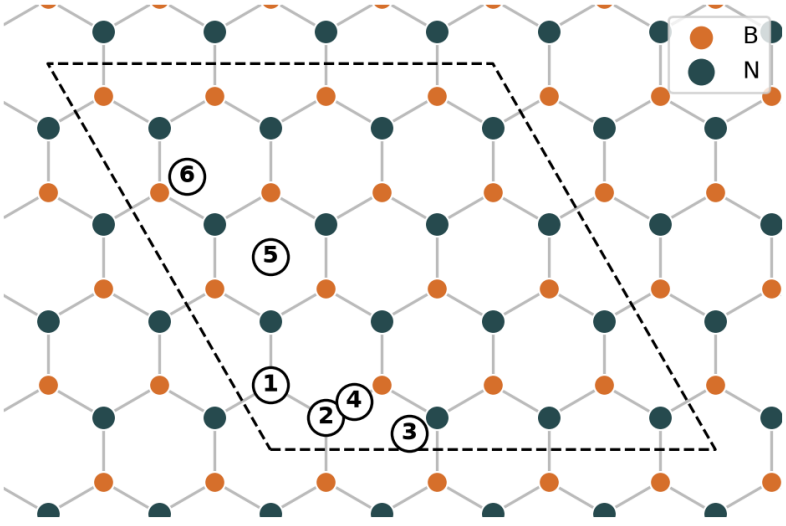
```
stb-adsorb -s structure.fdf -c calc.fdf -p dojo \
  --adsorbate H --site-type all --all-sites \
  --height 1.5 -O adsorption_run --save-report
```

Site search (pymatgen)

Type	Raw	Distinct
ontop	32	2
bridge	96	4
hollow	0	0
Total	128	**6**

Also written

`config_extra.fdf` per folder (spin, D3, dipole, fixed cell), `site_positions.dat`, `adsorption_sites.png`



1: on B 2: on N 4: B–N bond 5: hexagon center

3, 6: between an atom and the center

Check the labels

- On a honeycomb, the hollow appears as "bridge" 5
- Read the map, not only the names

stb-adsorbBsse: Ghost Folders at the Relaxed Geometry

```
stb-adsorbBsse --dir adsorption_run --save-report
```

What it does

- Reads each finished `sites/site_*/siesta.XV`
- Flags sites whose residual force is above tolerance
- Writes `bsse/site_*/bsse_slab/` and `bsse_adsorbate/`
- Inherits spin, D3, and dipole settings from the site

h-BN report

Site	F_{max} (eV/Å)	Status
site_1_ontop	0.005	ready
site_2_ontop	0.008	ready
site_3_bridge	0.010	ready
site_4_bridge	0.009	ready
site_5_bridge	0.007	ready
site_6_bridge	0.016	ready

6 pairs $\longrightarrow$ 12 single points (about 2 min each, 4 cores)

site_6 stopped at the 150-step limit: still below the 0.05 eV/Å warning threshold.

`stb-adsorbAnalysis`: Energies, BSSE, and Ranking

```
stb-adsorbAnalysis --dir adsorption_run \
  --save-report \
  --apply best_adsorbed_structure.fdf
```

What it reports

- E_{ads} per site, uncorrected and BSSE-corrected
- Residual-force check per site
- Ranking plot and `adsorption_curve.dat`
- `--apply`: copies the best site's structure

h-BN: E_{ads} per site (eV, vs. atomic H)

Site	Final state	raw	CP (STB)	CP + def.
site_1	B–H	−0.093	**−0.437**	−0.007
site_4	B–H	−0.104	−0.436	≈ site_1
site_3	phys.	−0.107	−0.034	**−0.049**
site_6	phys.	−0.103	−0.033	
site_5	phys.	−0.093	−0.033	
site_2	phys.	−0.079	−0.029	

Deformation energy

For B–H, the slab relaxes (B moves 0.32 Å out of plane), which costs +0.43 eV. **CP (STB)** computes $E_{\text{site}} - E_{\text{slab}}^{[\text{ghost}]} - E_{\text{H}}^{[\text{ghost}]}$ with the *deformed* slab, so this cost is lost. Adding it back, physisorption wins.

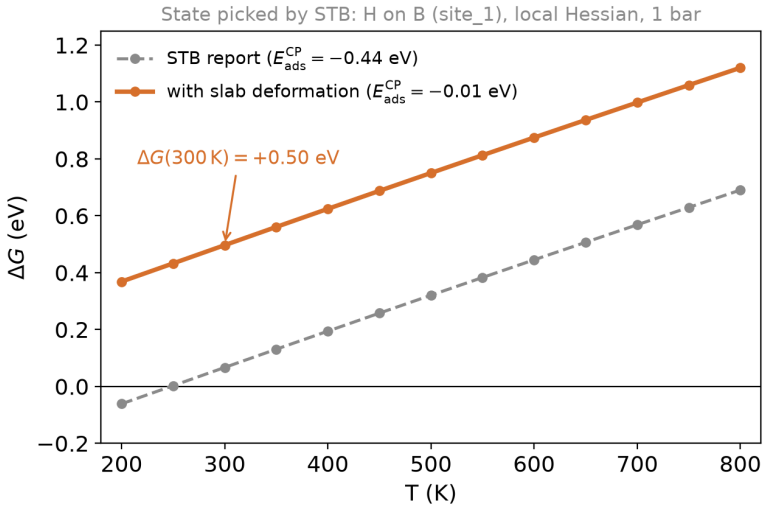
Stage 4: stb-adsorbGibbs

```
# Stage 3b: Hessian folders for the best site
stb-adsorbAnalysis --dir adsorption_run \
  --compute-gibbs --zpe-mode local
# run SIESTA in adsorption_run/gibbs/*/disp_*/
stb-adsorbGibbs --dir adsorption_run \
  --tmin 200 --tmax 800 --tstep 50 --save-report
```

Options

- `local`: only the adsorbate moves; 6 displacements per atom (6 SIESTA runs for H)
- `full`: Phonopy on site and clean slab; $6 \times 3N$ runs
- `--displacement` (default 0.015 Å)

Reports D_0, ΔZPE, ΔS (J/(mol K)), $\Delta G(T)$, T_d, and a rough desorption time.

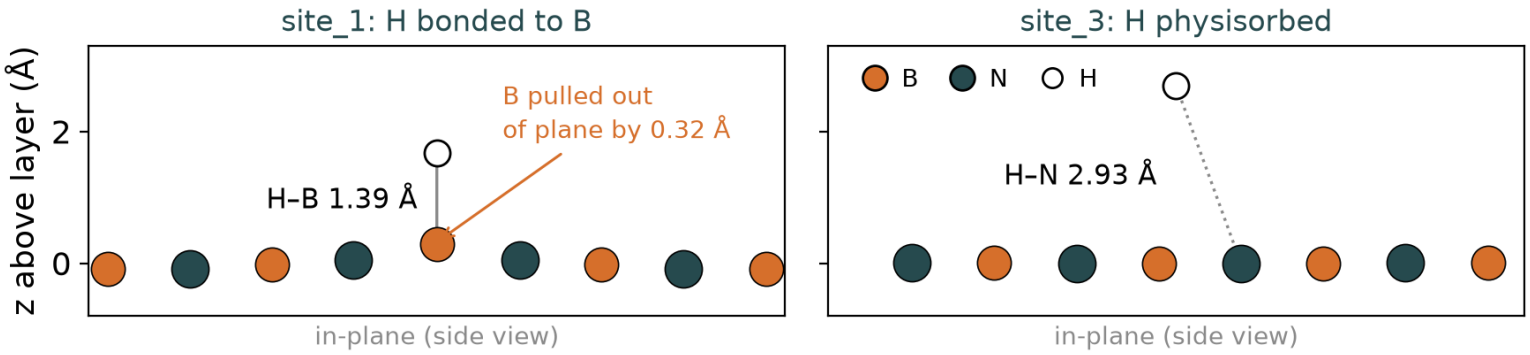


H modes on B	650, 650, 1224 cm^{-1} (no imaginary)
ΔZPE	+0.16 eV
ΔS (300 K)	−112 J/(mol K)
$-T\Delta S$ (300 K)	+0.35 eV

B–H is not stable on pristine h-BN

- With slab deformation: $\Delta G > 0$ at all T
- Relative to $\frac{1}{2}H_2$: every site $\approx +2$ eV

Reading the Result: Where Did H Go?



Checklist on h-BN

- **Final vs start:** site_4 (B–N bridge) moved onto B, same state as site_1; the others went to ∼2.7 Å
- **Spin:** 1 μ_B in every site, so it does not separate the two states
- **BSSE:** +0.06 to +0.09 eV, as large as E_{ads} itself

Two basins, almost degenerate

Chemisorbed B–H and physisorbed H differ by only ∼0.04 eV. site_1 and site_4 (same structure) differ by 11 meV: that is the numerical noise level.

Hands-on: H Adsorption on h-BN

Elastic Constants: Symmetry at Work

Hooke's Law in Voigt Notation

$$\sigma_i = \sum_{j=1}^{6} C_{ij}\,\gamma_j$$

$$\gamma = (\varepsilon_{xx}, \varepsilon_{yy}, \varepsilon_{zz}, 2\varepsilon_{yz}, 2\varepsilon_{xz}, 2\varepsilon_{xy})$$

1	2	3	4	5	6
xx	yy	zz	yz	xz	xy

C is a symmetric 6×6 matrix: at most **21** independent constants.

Units depend on dimensionality

3D: GPa 2D: N/m ($\sigma \cdot L_z$, independent of the vacuum) 1D: nN

Stress method (default)

Strain one component, read SIESTA's stress tensor: one series gives a whole **column** of C.

$$C_{ij} = \partial\sigma_i / \partial\gamma_j$$

Energy method

$$E = E_0 + \frac{V_0}{2}\,\gamma^T C\,\gamma$$

One curvature per pattern: off-diagonal terms need combined patterns such as $xx{+}yy$.

Neumann's Principle: Symmetry Fixes the Pattern of C

C must be invariant under every operation R of the point group:

$$C'_{ijkl} = R_{ia} R_{jb} R_{kc} R_{ld}\, C_{abcd} = C_{ijkl}$$

GaAs, zincblende ($\bar{4}3m$): 3 constants

$$\begin{pmatrix} C_{11} & C_{12} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{11} & C_{12} & 0 & 0 & 0 \\ C_{12} & C_{12} & C_{11} & 0 & 0 & 0 \\ 0 & 0 & 0 & C_{44} & 0 & 0 \\ 0 & 0 & 0 & 0 & C_{44} & 0 \\ 0 & 0 & 0 & 0 & 0 & C_{44} \end{pmatrix}$$

h-BN monolayer ($\bar{6}m2$), in plane

$$\begin{pmatrix} C_{11} & C_{12} & 0 \\ C_{12} & C_{11} & 0 \\ 0 & 0 & \frac{C_{11}-C_{12}}{2} \end{pmatrix}$$

2 in-plane constants (5 for the full 3D tensor)

Fewer constants, fewer calculations

- Triclinic: 21 constants, 6 strain series
- Cubic: 3 constants, 2 series Hexagonal 2D: 1 series

How STB Uses the Point Group

Point group from pymatgen (`SpacegroupAnalyzer`), operations R in **Cartesian** coordinates.

`--symmetry-method basic`

Pairwise test on the strain tensors:

$$R\,\varepsilon^{(i)}\,R^{T} = \pm\,\varepsilon^{(j)}$$

If true, direction j is skipped and its stress is rebuilt:

$$\sigma^{(j)} = \pm\,R\,\sigma^{(i)}\,R^{T}$$

Cubic: $\{xx, yy, zz\}$, $\{yz, xz, xy\}$. Hexagonal: a 60° rotation sends ε_{xx} to a *mixed* tensor, so $C_{11} = C_{22}$ is **missed**.

`--symmetry-method full`

Write C as 21 unknowns c; each R gives a linear map L_R:

$$(L_R - I)\,c = 0 \quad \forall R$$

SVD null space $\rightarrow$ basis $\{B_k\}$, one per independent constant.

$$\sigma = \sum_k x_k\,B_k\,\gamma$$

Directions chosen greedily until the design matrix reaches rank K; one global least-squares fit for all x_k.

Source: `stb/core/symmetry.py` (`equivalent_strain_modes`, `symmetry_allowed_basis`, `select_directions_by_rank`, `fit_stiffness_matrix`).

Symmetry Lives in the Cartesian Frame

Voigt strains are applied along Cartesian x, y, z. The reduction works only if these axes match the crystal axes.

Same GaAs, two cells (Stage 1 only)

Cell	`basic`	`full`
cubic axes	2 of 6	2 of 6
Part 1 cell (rotated)	6 of 6	1 of 6

Both detect $\bar{4}3m$, but in the rotated frame C has extra nonzero entries: C_{11} there is *not* the textbook C_{11}.

Standard zincblende cell

$\mathbf{a}_1 = \frac{a}{2}(0,1,1)$, $\mathbf{a}_2 = \frac{a}{2}(1,0,1)$, $\mathbf{a}_3 = \frac{a}{2}(1,1,0)$

Ga $(0,0,0)$, As $(\frac{1}{4}, \frac{1}{4}, \frac{1}{4})$

Before Stage 1

Check that the reported point group is the expected one ($\bar{4}3m$, $\bar{6}m2$) and that the cell is in the standard orientation. Do not raise `--symprec` to hide a distorted cell.

Clamped Ion vs Relaxed Ion: Symmetry Decides

Internal strain

With 2+ atoms per cell, a strain can push the sublattices against each other. STB forces `MD.Steps 0`: ions stay at the scaled positions $\Rightarrow$ **clamped-ion** C_{ij}.

When does it happen?

Strain	GaAs	h-BN
xx	$F = 0.0001$	$F = 0.13$
xy	$F = 0.13$	$F = 0.28$

Max force (eV/Å) at 1% strain. Only strain with the symmetry of the sublattice mode couples: T_2 (shear) in T_d; E' ($\varepsilon_{xx} - \varepsilon_{yy}$, ε_{xy}) in D_{3h}. h-BN biaxial strain (A_1'): $F = 5 \times 10^{-5}$.

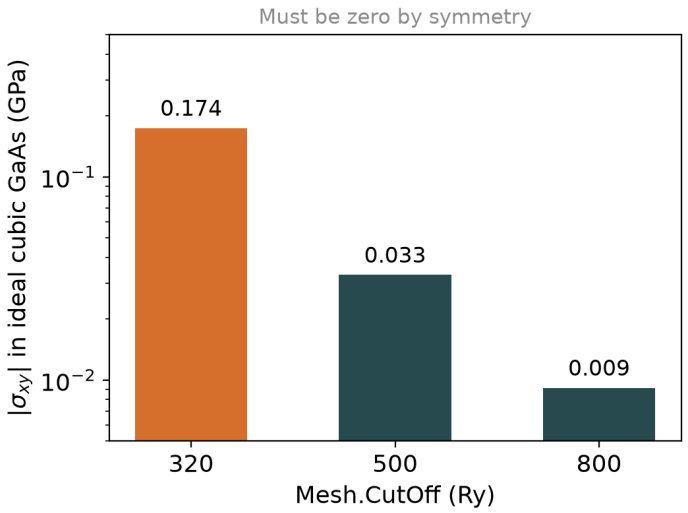
Relaxing the ions (cell fixed)

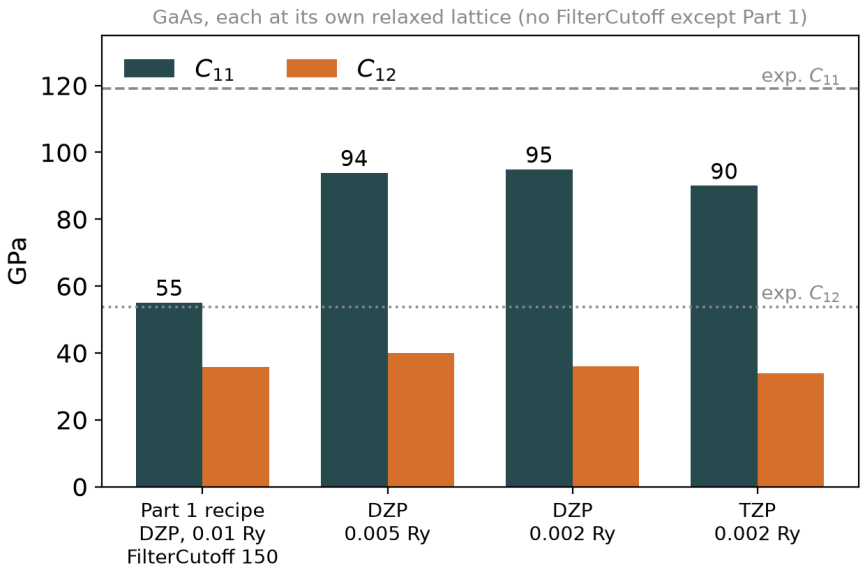
	clamped	relaxed
GaAs C_{11} (GPa)	94.1	94.1
GaAs C_{44} (GPa)	64.2	**47.5**
h-BN C_{11} (N/m)	307.3	**294.2**
h-BN C_{12} (N/m)	54.2	**67.4**
h-BN $C_{11} + C_{12}$	361.5	361.6

Compare with experiment: relaxed ion

STB has no relaxed-ion option: here `config_extra.fdf` was edited by hand (`MD.Steps 100`).

Elastic Constants Need More Convergence





Egg-box: the real-space grid of the FCC cell has only trigonal symmetry, so it creates a shear stress that $\bar{4}3m$ forbids.

Reference structure

`MD.MaxStressTol 0.01 GPa` (default: 1 GPa). The Part 1 GaAs cell kept ∼1 GPa of residual stress.

Part 1 recipe: C_{11} 40% too low

- `FilterCutoff` + `EnergyShift 0.01 Ry` soften Ga 3d
- Used here: DZP, 0.005 Ry, no filter, 500 Ry

Mechanical Stability and Derived Moduli

Born stability: C positive definite

Cubic:

$C_{11} - C_{12} > 0,\ C_{11} + 2C_{12} > 0,\ C_{44} > 0$

2D: $C_{11} > 0,\ C_{66} > 0,\ C_{11}C_{22} > C_{12}^2$

STB (3D): all eigenvalues of C positive; works for any point group.

2D moduli

$$Y_{2D} = \frac{C_{11}C_{22} - C_{12}^2}{C_{22}}, \qquad \nu = \frac{C_{12}}{C_{22}}$$

3D moduli: Voigt–Reuss–Hill

$B_V = \frac{1}{9}\sum_{i,j\le 3} C_{ij}$ $\qquad$ $B_R = \left(\sum_{i,j\le 3} S_{ij}\right)^{-1}$

$B_H = \frac{1}{2}(B_V + B_R)$, same for G; then

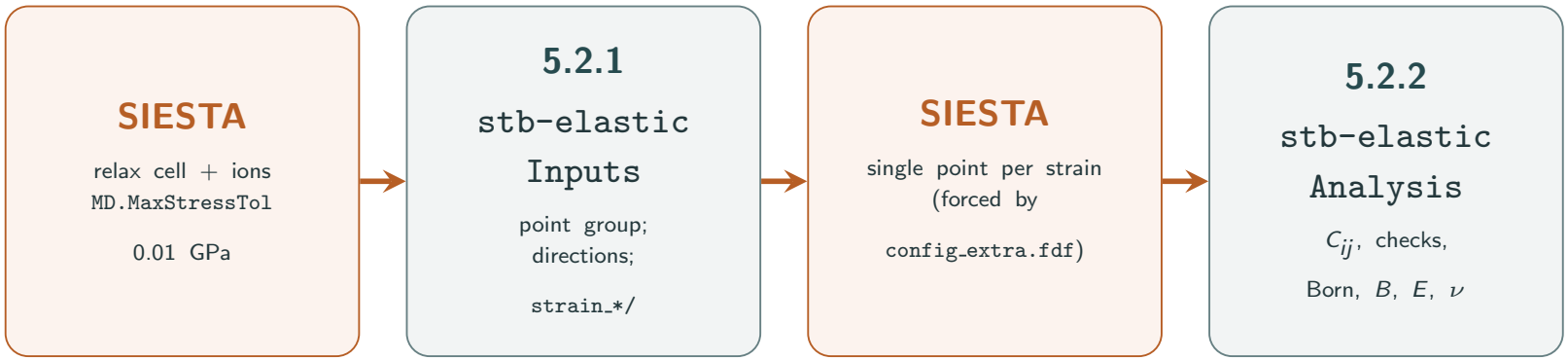
$$E = \frac{9BG}{3B+G}, \qquad \nu = \frac{3B-2G}{2(3B+G)}$$

$S = C^{-1}$. Cubic: $B = (C_{11} + 2C_{12})/3$.

Anisotropy

$A^U = 5G_V/G_R + B_V/B_R - 6$ $\quad$ (0 = isotropic)

Workflow 5.2: Elastic Constants with STB



Choices made in Stage 1

`--method stress|energy`
`--symmetry-method basic|full`
`--max 1.0 --steps 5` ($\pm 1\%$, 5 points)

Must match

- Stage 2: same `--method` and `--symmetry-method`
- Point group from `reference_structure.fdf` (undeformed cell)

stb-elasticInputs: Symmetry Picks the Directions

```
stb-elasticInputs -s structure.fdf -c calc.fdf --dirs all \
  --symmetry-method full --max 1.0 --steps 5 -o elastic_stress_full --save-report
```

GaAs: point group $\bar{4}3m$, 24 operations

Dir.	basic	full
xx	RUN	RUN
yy, *zz*	from *xx*	fit
xy	RUN	RUN
xz, *yz*	from *xy*	fit
runs	10	10

energy: *xx*, *yz*, *xx*+*yy* (15 runs)

h-BN: point group $\bar{6}m2$, 12 operations

Dir.	basic	full
xx	RUN	RUN
yy	RUN	fit
xy	RUN	fit
zz, *xz*, *yz*	skipped: vacuum	
runs	15	**5**

energy: *xx*, *xy* (10 runs)

Every folder gets `config_extra.fdf` (`MD.Steps 0`, `MD.VariableCell false`) `%included` at the *top* of `calc.fdf`: first occurrence wins.

stb-elasticAnalysis: GaAs

```
stb-elasticAnalysis -d elastic_stress_full --symmetry-method full --save-report --save-gnuplot
```

Four routes, one tensor (GPa)

Run	runs	C_{11}	C_{12}	C_{44}
stress, `basic`	10	94.1	40.0	64.2
stress, `full`	10	94.1	40.0	64.2
stress, 6 explicit	30	94.1	40.0	64.2
energy	15	94.6	39.4	64.7
relaxed ion	10	94.1	40.0	**47.5**
Exp. (RT)		118.8	53.8	59.4

Relaxed ion: $B = 58$ GPa, $E = 93$ GPa, $\nu = 0.23$, Zener $2C_{44}/(C_{11} - C_{12}) = 1.76$ (exp. 1.83). STABLE.

Exp.: Bateman, McSkimin & Whelan, J. Appl. Phys. 30, 544 (1959).

[3] Quality checks

Egg-box (stress vs energy):

xx 0.55%, xy 0.71%	OK
Fit: $R^2 \geq 0.996$	OK
max $\lvert C_{ij} - C_{ji} \rvert$ (6 explicit): 0.04 GPa	OK

Checks catch noise, not a bad model

The Part 1 recipe ($C_{11} = 55$ GPa) also passes every check.

stb-elasticAnalysis: h-BN

In-plane constants (N/m)

Run	runs	C_{11}	C_{22}	C_{12}	C_{66}
stress, basic	15	307.3	307.0	54.2	126.5
stress, full	5	307.3	307.3	54.1	126.6
energy	10	307.9	307.9	54.5	126.7
relaxed ion	15	294.2	293.8	67.4	113.2

Relaxed ion: $Y_{2D} = 279$ N/m, $\nu = 0.23$. The sublattice shift (also the origin of piezoelectricity in h-BN) lowers $C_{11} - C_{12}$ only [Michel & Verberck, PRB 80, 224301 (2009)].

What basic buys

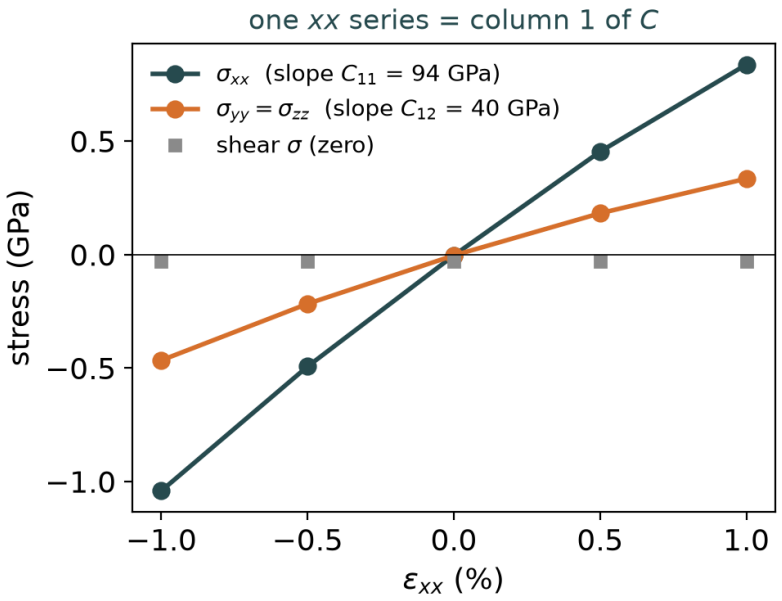
C_{66} measured: 126.5
$(C_{11} - C_{12})/2$: 126.5
C_{12} from xx vs yy: $\Delta = 0.27$

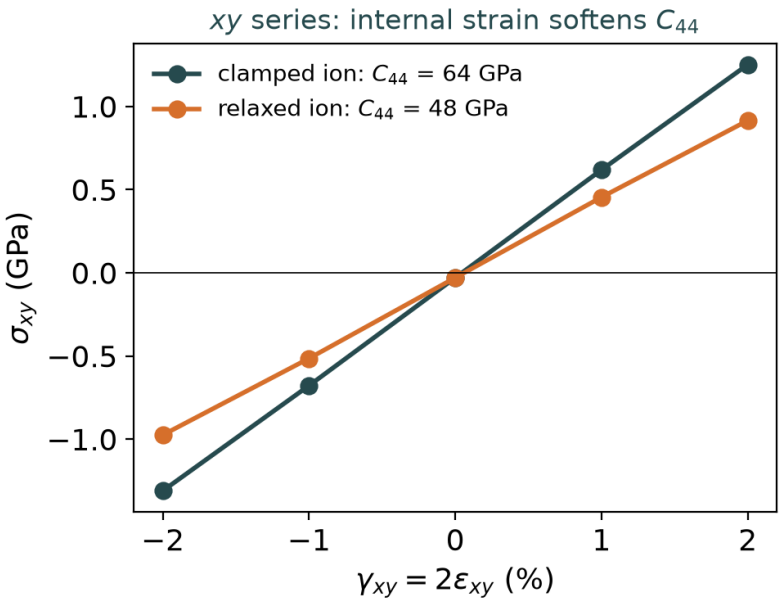
In full, all three are true by construction, so they test nothing.

Cost

full: 3× cheaper, same numbers.

Reading the Fits





Checklist

- Linear within $\pm 1\%$ (R^2)
- Zero-strain stress ≈ 0 (relaxed reference)
- Forbidden components stay ≈ 0

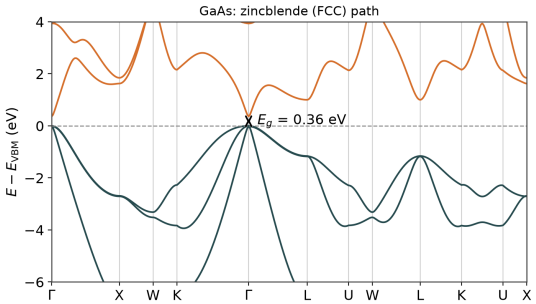
Files

`elastic_stage2.txt`, `elastic_plots/*_fit.dat` (columns: strain, γ, six σ components)

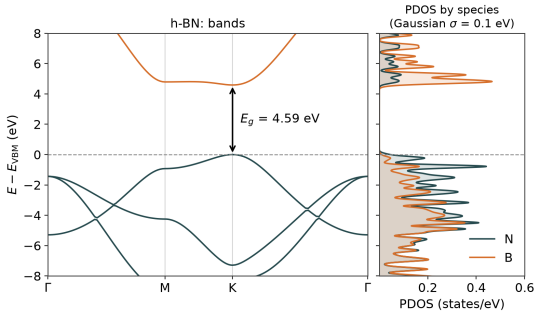
Hands-on: Elastic Constants of GaAs and h-BN

Closing

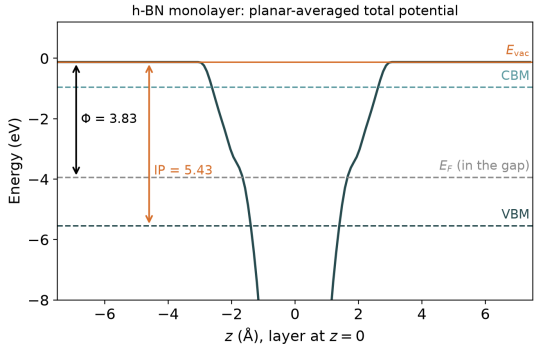
Part 2 in Six Figures



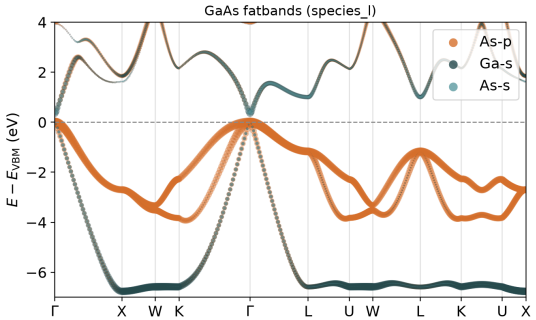
Bands: FCC path, PBE gap



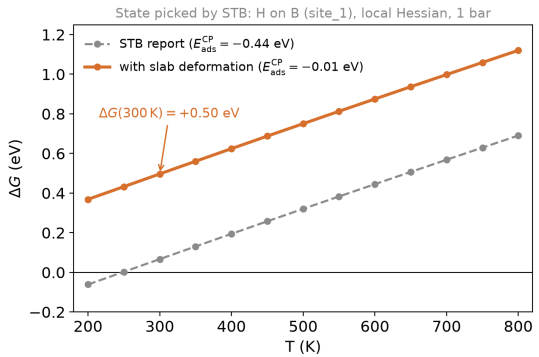
Bands + PDOS of h-BN



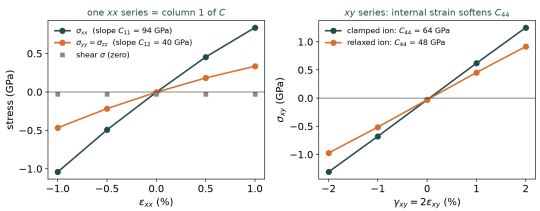
Work function, E_F mid-gap



Fatbands: orbital character



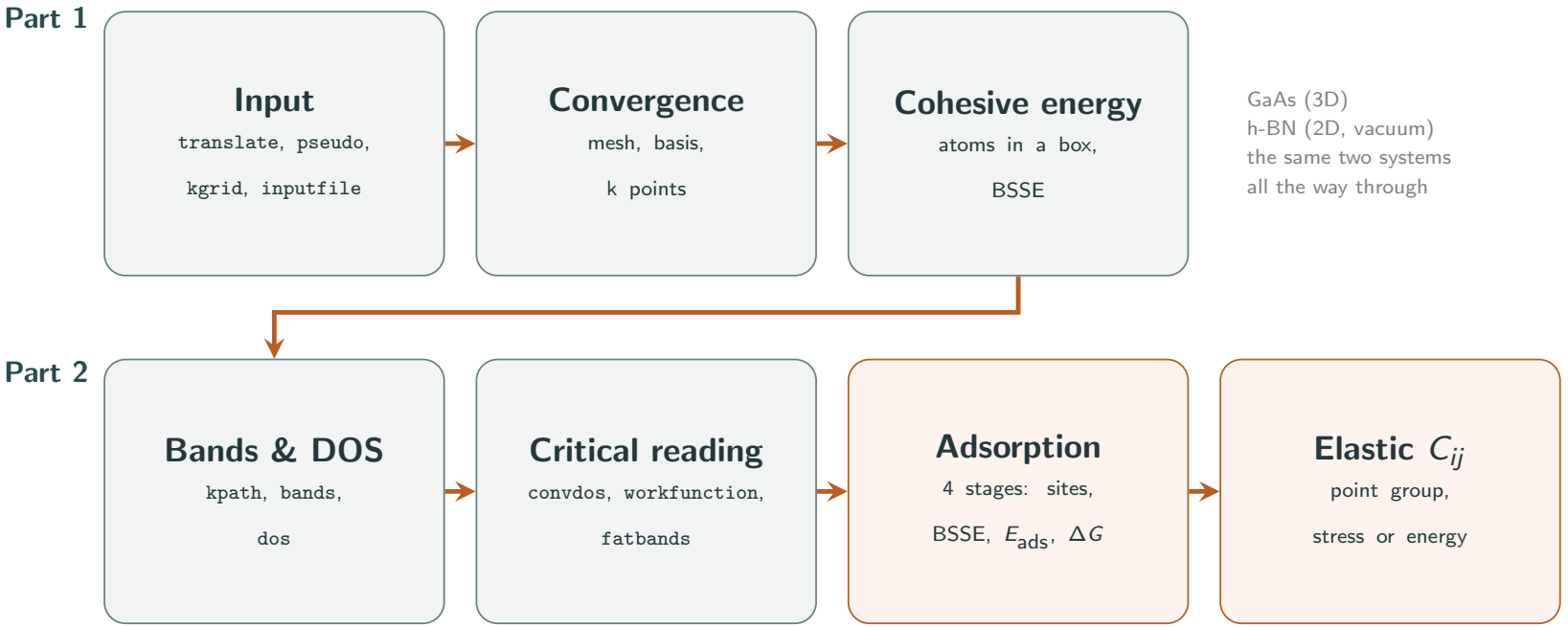
H on h-BN: $\Delta G > 0$



C_{ij}: symmetry + internal strain

Every figure came from a SIESTA output read by one STB command.

The Course in One Map



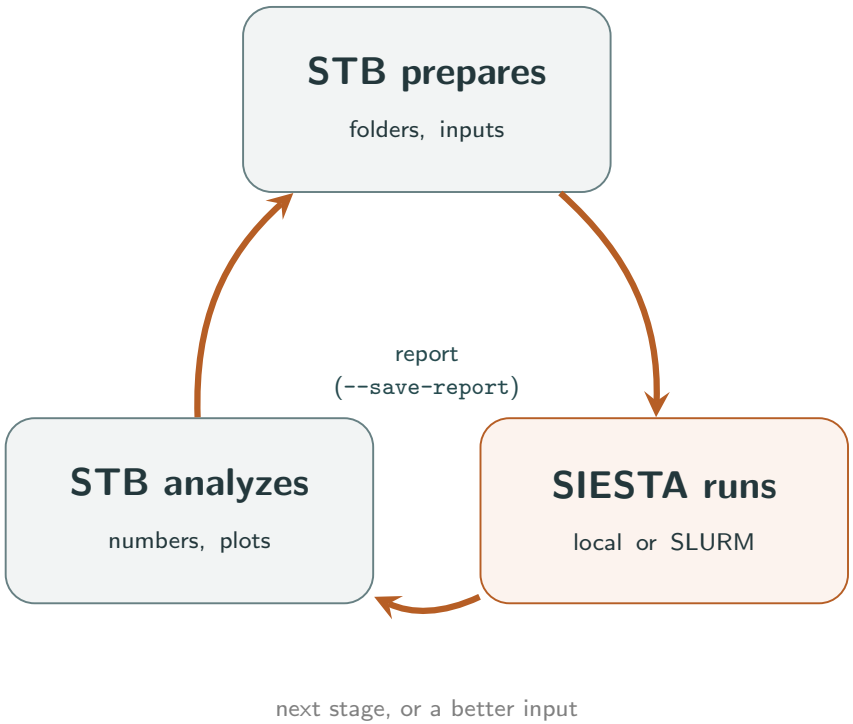
Single tools

One output → one property (bands, DOS, work function, fatbands).

Workflows (orange)

Several SIESTA rounds; each stage reads the previous one and writes the next folders.

One Pattern for Every Property



What makes it reproducible

- The same `calc.fdf` for every reference
- Forced settings in `config_extra.fdf`, `%included` first
- A report next to every result
- `run/` to work, `completed/` to compare

STB never runs SIESTA

The boundary is explicit: you decide where and how the calculations run.

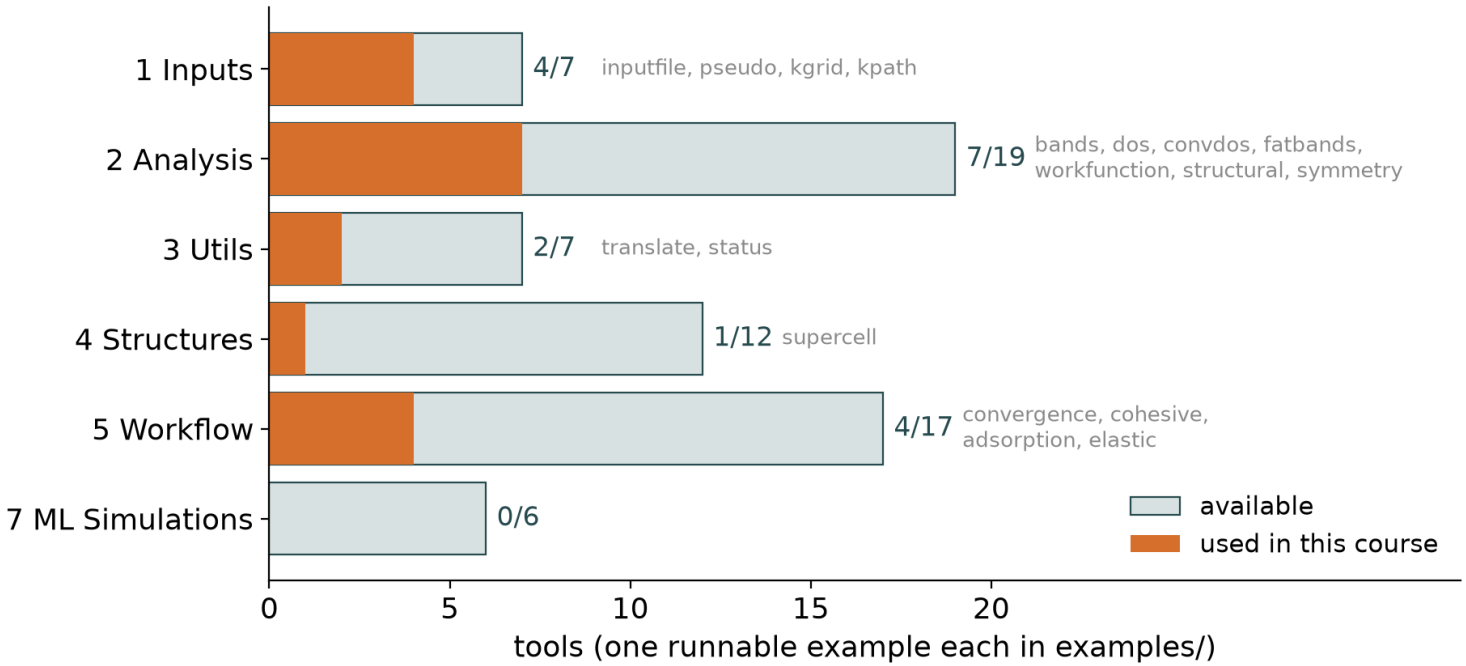
Before You Trust a Number

Check	What happened here
Converged for *this* property	Part 1 basis (tested on energy): GaAs C_{11} 40% too low
Reference really at rest	default `MD.MaxStressTol` left $\sim$1 GPa in GaAs
Symmetry and orientation	GaAs read as rhombohedral (`kpath -p 0.01`); rotated cell breaks the C_{ij} reduction
Label vs final geometry	site_4 "bridge" relaxed onto B; "bridge 5" is the hollow
What a correction contains	BSSE without slab deformation: -0.44 instead of -0.01 eV
Noise floor	11 meV between identical sites; egg-box shear stress
Limits of the model	PBE gap 0.36 vs 1.42 eV; clamped vs relaxed ion; harmonic ΔS

Quality checks catch noise, not a wrong model

- Every STB check passed with the unconverged GaAs basis
- Compare with a reference value before you report

What We Used, and What Is Left



Natural next steps: `5.1` strain, `5.15` equation of state, `5.4` phonons, `5.9` NEB, `2.6` Bader, `2.14` COOP.

Fast preview: `7.4 stb-mlelastic`, `7.6 stb-mladsorb` with a MACE potential before spending DFT time.

Where to Find Everything

STB-Suite

Docs: `stb-suite.github.io/stb`
Code: `github.com/stb-suite/stb`

```
stb-suite        # menu: type a code, e.g. 5.2.1
stb-<tool> --help
examples/<code>-<tool>/README.md
```

Every tool has a guide and a runnable example script.

Course material

```
aula1/
  relaxation/  convergence/
  cohesive_energy/
aula2/
  bandas_dos/
  fatbands_workfunction/
  adsorption/
  elastic/
```

Same layout in every example

- `run/` or `input/`: start here
- `siesta_stage*_complete/`: SIESTA done, ready for the next STB stage
- `complete/`: final results, to compare with yours

Questions?

prepare $\rightarrow$ run $\rightarrow$ analyze $\rightarrow$ question the number

Acknowledgments

First, I would like to thank the **SEEDMOL COMMITE**, for the invitation.

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CAPES

Laboratório de Computação em Ciência de Materiais (LCCMat)



Thank You & Let's Connect!

We're always open to new collaborations...

...especially if they involve discussing physics over a good cup of coffee!



my email:
carlos.bastos@unb.br

Thank you!